
Volatility Managed Portfolios

A Critical Assessment Across Five Asset Classes

January 2000 – December 2025

Advanced Master Project

EDHEC Business School

Volatility Management Strategies

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Abstract

This report critically evaluates volatility-targeting (VT) strategies applied to five asset classes — US equities, US long-term government bonds, commodities, a currency pair, and a long-short equity momentum factor — over the period January 2000 to December 2025 (6,542 daily observations). We implement the strategy using four volatility forecasting models: one-month realised volatility (RV 1M), six-month realised volatility (RV 6M), a GJR-GARCH(1,1) model with Student- t errors, and an EGARCH(1,1) model. Target volatility is computed via a fully expanding window to avoid look-ahead bias, following [Bongaerts et al. \(2020\)](#).

Our findings are heterogeneous across asset classes. For US equities (S&P 500 TR), GJR-GARCH raises the Sharpe ratio from 0.658 to 0.785 while reducing the maximum drawdown from -50.9% to -36.8% at a transaction cost of 10 bps. The momentum factor benefits most substantially ($\Delta\text{Sharpe} = +0.297$). Conversely, volatility management destroys value for commodities ($\Delta\text{Sharpe} = -0.073$) and fails to improve bonds materially. We show that the breakeven transaction cost for the S&P 500 is approximately 63.6 bps, well above typical institutional costs. Leverage constraints and turnover-reduction techniques (threshold banding, EMA smoothing) are analysed: EMA smoothing reduces annual turnover by 47% for equities with minimal Sharpe sacrifice. NBER business cycle analysis confirms that VM strategies provide meaningful recession protection for equities and momentum but partially erode the flight-to-quality benefit of bonds during contractions.

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1. Theory of Volatility-Targeting Strategies

1.1. Conceptual Foundation

Volatility-targeting (VT) strategies exploit a well-established empirical regularity: whereas expected returns are notoriously difficult to forecast, *conditional volatility is predictable* at both short and medium horizons (Andersen et al., 2003). This predictability creates an economically attractive opportunity: an investor who reduces exposure before high-volatility regimes — when the variance-risk premium is elevated and crash risk is greatest — should obtain superior risk-adjusted returns without relying on any return-forecasting ability.

The foundational modern treatment is Moreira & Muir (2017), who demonstrate that scaling a portfolio's weight inversely with its predicted variance generates portfolios with higher Sharpe ratios and substantially reduced maximum drawdowns across US equity factors. Harvey et al. (2018) extend this result to a broader cross-section of asset classes and document consistent improvements in risk-adjusted returns under institutional-quality transaction cost assumptions. CTAs, hedge funds, and systematic macro strategies have widely adopted volatility targeting; assets under management in related strategies (risk-parity funds, variable annuities, systematic trend-following) exceed \$1 trillion.

1.2. Formal Framework

At the beginning of month t , the weight on asset i is:

$$w_{i,t} = \frac{\sigma^{\text{target}}}{\hat{\sigma}_{i,t}}, \quad (1)$$

where σ^{target} is a long-run target volatility level and $\hat{\sigma}_{i,t}$ is the one-period ahead conditional volatility forecast. The managed-portfolio return is:

$$r_t^{\text{vm}} = \begin{cases} r_{f,t} + w_{i,t}(r_{i,t} - r_{f,t}) & \text{long-only assets,} \\ w_{i,t} r_{i,t} & \text{long-short (self-financing) assets.} \end{cases} \quad (2)$$

When predicted volatility is below target, the strategy levers up, exploiting the low-volatility environment; when predicted volatility rises, the strategy scales down, limiting drawdown risk. The constant $k = 1$ (no ex-post re-scaling) is used throughout this study, so that the strategy is implementable in real time. An optional maximum leverage constraint $w_{i,t} \leq w_{\text{max}}$ prevents unbounded leverage in extremely low-volatility regimes.

1.3. Look-Ahead Bias in Target Volatility

A subtle but critical implementation issue concerns the computation of σ^{target} . Using the full-sample standard deviation — as in [Moreira & Muir \(2017\)](#) and [Harvey et al. \(2018\)](#) — introduces *look-ahead bias* by implicitly assuming that the portfolio can be rescaled using future information. [Liu et al. \(2019\)](#) demonstrate that Sharpe ratio improvements largely vanish once this bias is removed for equity factor portfolios.

Following [Bongaerts et al. \(2020\)](#) (p. 3), we eliminate look-ahead bias by computing σ^{target} as an *expanding-window* standard deviation:

$$\sigma_t^{\text{target}} = \text{std}(r_1, r_2, \dots, r_{\tau_{t-1}}), \quad (3)$$

where τ_{t-1} denotes the last trading day of month $t - 1$. This ensures that σ_t^{target} is computable at time t without any future data. Under this definition, σ_t^{target} converges monotonically to the full-sample standard deviation as $t \rightarrow T$, but is strictly backward-looking at each point in time. We impose a minimum 48-month estimation window before computing portfolio weights to ensure statistical reliability of the initial target volatility estimates.

1.4. Theoretical Mechanisms and Conditions for Profitability

Table 1 summarises the main theoretical channels through which volatility-targeting is expected to generate value.

Table 1: Theoretical Mechanisms Supporting Volatility-Targeting Strategies

Mechanism	Description	Reference
Variance-risk premium	High vol states coincide with elevated VRP; de-risking avoids paying the premium while benefiting from low-vol upside	Moreira & Muir (2017)
Inverse vol–return relationship	Conditional Sharpe ratios are systematically lower in high-volatility regimes across most asset classes	Harvey et al. (2018)
Drawdown mitigation	Automatic de-risking before crises reduces left-tail exposure and limits momentum crashes	Bongaerts et al. (2020)
No return-timing required	Strategy exploits only vol predictability, which is robust, without requiring expected-return forecasts	Harvey et al. (2018)
Reduced negative skewness	Scaling down before vol spikes truncates the left tail, improving higher moments of the return distribution	Moreira & Muir (2017)

The strategy is expected to add value under three conditions: (1) volatility is sufficiently predictable ($R^2 > 0$); (2) the conditional Sharpe ratio is decreasing in volatility (i.e., risk premia do not adequately compensate for spikes in risk); and (3) transaction costs are sufficiently low relative to the frequency and magnitude of weight changes. If volatility is unpredictable or if the conditional Sharpe ratio is *positively* correlated with volatility — as is plausibly the case for government bonds during flight-to-quality episodes — the strategy can destroy value.

1.5. Critiques and Limitations of the Literature

[Liu et al. \(2019\)](#) identify three key challenges: (1) look-ahead bias in σ^{target} inflates results; (2) performance is concentrated in the GFC period and may not persist out of sample; and (3) the strategy is primarily a crisis-avoidance mechanism rather than a source of structural alpha. The finding of [Liu et al. \(2019\)](#) that Sharpe ratio improvements are statistically insignificant after bias correction is particularly sobering and motivates our bias-free implementation. Our empirical analysis revisits these claims across a broader asset universe and a longer post-crisis period (2010–2025).

2. Data

2.1. Asset Universe and Sources

The study covers five assets representing distinct risk premia and asset classes. All data were sourced from Bloomberg and Kenneth French's data library, spanning **January 2000 to December 2025** (6,542 daily observations, 312 monthly observations after the 48-month warm-up period).

Table 2: Asset Universe: Description and Sources

Ticker	Asset	Description	Return Type
SPXT	S&P 500 TR	S&P 500 Total Return Index; dividends reinvested	Total Return
LT09TRUU	US LT Bonds TR	Bloomberg US Long Treasury TR Index; coupon reinvested	Total Return
BCOMCOT	Bloomberg Commodities	Bloomberg Commodity Total Return Index; roll yield and collateral return included	Total Return
USDJPY	USD/JPY	US Dollar / Japanese Yen spot exchange rate	Price Return
MOM	US Momentum Factor	Fama-French US momentum factor (long winners, short losers); sourced from Kenneth French's data library	L/S (decimal)
USGG3M	3-Month US T-Bill	Annualised yield; used as risk-free rate proxy	Yield

Return construction. Daily simple returns are computed as $r_t = P_t/P_{t-1} - 1$ for the four index or rate series. The MOM factor is expressed as a daily percentage return and converted to decimal by dividing by 100. The risk-free rate is compounded daily as $r_{f,t} = (1 + y_t/100)^{1/252} - 1$, where y_t is the annualised 3-month T-bill yield. Monthly returns are the cumulative product of daily returns within each calendar month.

Total return consideration. A key design choice is using total return indices for equities, bonds, and commodities, which include dividends, coupon income, and roll/collateral returns respectively. This is essential for fair comparison: the volatility-targeting weight applies to the complete economic return, not merely the price change. For USD/JPY (a currency rate), only price return is available; the carry differential is not explicitly modelled, constituting a conservative treatment of FX exposure.

2.2. Descriptive Statistics

Table 3 reports annualised descriptive statistics for the full sample.

Table 3: Descriptive Statistics: January 2000 – December 2025

Asset	Ann. Return	Ann. Volatility	Sharpe	Skewness	Exc. Kurtosis
S&P 500 TR	7.8%	19.4%	0.31	−0.35	10.63
US LT Bonds TR	4.2%	6.6%	0.37	−0.01	2.64
Bloomberg Commodities	6.6%	34.0%	0.14	−0.69	9.38
USD/JPY	1.3%	9.7%	−0.04	−0.13	4.27
US Momentum Factor	3.8%	16.7%	0.12	−0.71	6.54

Several features are noteworthy. First, commodities exhibit the highest volatility (34.0%) but the lowest Sharpe ratio among directional assets, reflecting difficulty in earning a structural risk premium. Second, all assets display negative skewness and excess kurtosis — consistent with fat-tailed financial return distributions — confirming the relevance of tail-risk management. Third, the momentum factor, despite being self-financing, exhibits pronounced negative skewness (−0.71) and kurtosis (6.54), capturing the well-documented momentum crash phenomenon (Moreira & Muir, 2017). USD/JPY returns a negative Sharpe ratio on a simple buy-and-hold basis, consistent with the well-known puzzle of non-zero excess returns for currency carry under risk-neutrality.

2.3. Correlation Structure

As shown in Figure 1, cross-asset return correlations are uniformly near zero (all $|\rho| \leq 0.02$), which justifies treating each asset independently in a single-asset volatility-targeting framework. This near-orthogonality also implies that multi-asset diversification benefits are largely independent of any intra-asset vol-timing benefit studied here, suggesting that the strategies studied could serve as building blocks in a multi-asset portfolio without introducing spurious cross-asset dependencies.

Asset Universe: Normalised Prices & Correlation Structure

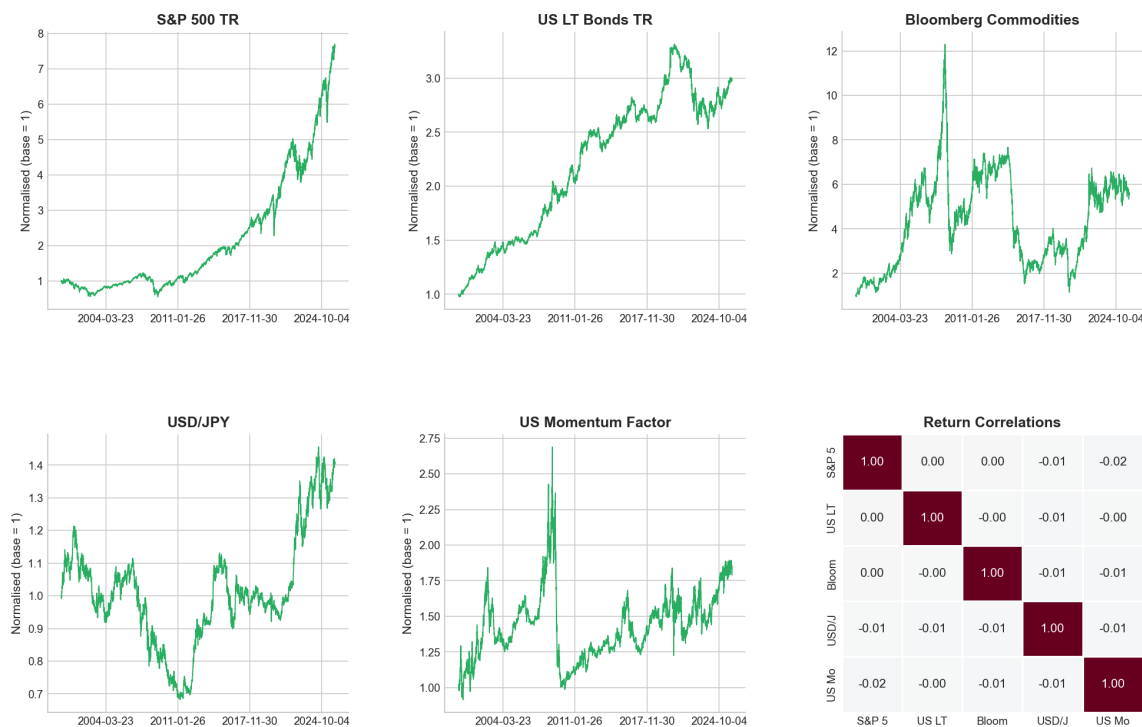


Figure 1: Asset Universe: Normalised Prices (base = 1) and Pairwise Return Correlations (January 2000 – December 2025). All cross-asset correlations are below 0.02 in absolute value. Note the dramatically different return trajectories: S&P 500 grows 7.6 $\times$ over the sample while Bloomberg Commodities peaks in 2008 and subsequently declines from a 12 $\times$ high to approximately 6 $\times$.

3. Methodology

3.1. Implementation Choices

Table 4 summarises the key design choices made in this study and their academic justification.

Table 4: Implementation Choices for the Volatility-Targeting Strategy

Parameter	Value / Rule	Rationale
Target volatility σ^{target}	Expanding-window std of daily re- turns up to end of $t - 1$	Avoids look-ahead bias (Bongaerts et al., 2020)
Scaling constant k	1 (no ex-post rescaling)	Implementable in real time
Leverage cap	$w_{\max} = 2.0 \times$ (baseline)	Realistic institutional constraint
Transaction cost	10 bps one-way (baseline)	Consistent with institutional estimates (Frazzini et al., 2018)
TC application	$r_t^{\text{net}} = r_t^{\text{vm}} - \Delta w_t \cdot c$	One-way cost on weight change
Return frequency	Monthly (compounded from daily)	Avoids monthly-return bias
Min. estimation window	48 months (≈ 4 years)	Ensures reliable GARCH estimation

3.2. Volatility Forecasting Models

Four forecasting models are implemented, all using strictly backward-looking (expanding window) data at the point of each monthly weight computation.

RV 1M. The one-month realised volatility is the equal-weighted standard deviation of daily returns in month $t - 1$, annualised by $\sqrt{252}$:

$$\hat{\sigma}_{i,t}^{\text{RV1M}} = \sqrt{\frac{252}{n_{t-1}} \sum_{s \in t-1} r_{i,s}^2}, \quad (4)$$

where n_{t-1} is the number of trading days in month $t - 1$. This is the benchmark model of [Moreira & Muir \(2017\)](#). Its parsimony makes it a natural baseline, but it reacts sharply to individual monthly outliers and can overestimate volatility for several months after a crisis.

RV 6M. The six-month realised volatility uses daily returns over months $t - 6$ to $t - 1$:

$$\hat{\sigma}_{i,t}^{\text{RV6M}} = \sqrt{\frac{252}{N_t} \sum_{s=\tau_{t-6}}^{\tau_{t-1}} r_{i,s}^2}. \quad (5)$$

The longer window reduces noise at the cost of slower reaction to regime shifts. The implicit exponential weighting implied by the longer window also means that crisis months are diluted over six months rather than one, reducing post-crisis overestimation at the expense of underestimating at the onset of a spike.

GJR-GARCH(1,1). The GJR-GARCH model of [Glosten et al. \(1993\)](#) captures the *leverage effect* — the well-documented asymmetric response of equity volatility to negative news —

which arises from financial leverage and loss-aversion mechanics. Estimated on daily returns scaled by 100:

$$r_{i,t} = \sigma_{i,t} \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim t_\nu, \quad (6)$$

$$\sigma_{i,t}^2 = \omega + \alpha r_{i,t-1}^2 + \gamma r_{i,t-1}^2 \mathbf{1}[r_{i,t-1} < 0] + \beta \sigma_{i,t-1}^2, \quad (7)$$

where $\gamma > 0$ captures the asymmetric effect. The persistence of shocks is $\alpha + \gamma/2 + \beta < 1$. The monthly forecast is the average 21-day-ahead variance under the GARCH recursion: $\hat{\sigma}_{i,t}^{\text{GARCH}} = \sqrt{252 \cdot \bar{V}_{21}}$. Student- t errors are used to accommodate the fat-tailed nature of daily returns evident in Table 3. The model is estimated recursively on all available data at the end of each month (expanding window) using quasi-maximum likelihood.

EGARCH(1,1). The EGARCH model of Nelson (1991) operates on log-variance, eliminating non-negativity constraints and allowing the sign of the return shock to affect conditional variance independently of its magnitude:

$$\log \sigma_{i,t}^2 = \omega + \beta \log \sigma_{i,t-1}^2 + \alpha \frac{|r_{i,t-1}|}{\sigma_{i,t-1}} + \gamma \frac{r_{i,t-1}}{\sigma_{i,t-1}}. \quad (8)$$

The log specification ensures $\sigma^2 > 0$ for all parameter values, providing greater numerical stability for assets like USD/JPY with near-zero drift. The EGARCH serves as a robustness check for the GJR-GARCH results; differences in portfolio performance between the two GARCH variants provide indirect evidence of the importance of the leverage specification.

3.3. Forecast Evaluation Methodology

Error metrics. We report Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) against the subsequent month's realised volatility (equal-weighted daily standard deviation over the next 21 trading days).

Mincer-Zarnowitz regression. The MZ regression (Mincer & Zarnowitz, 1969) assesses forecast unbiasedness and efficiency:

$$\sigma_{i,t+1}^{\text{realised}} = \alpha + \beta \hat{\sigma}_{i,t} + u_{t+1}. \quad (9)$$

An unbiased and efficient forecast satisfies $\alpha = 0$ and $\beta = 1$ jointly. We test this via an F -test (MZ joint test); rejection ($p < 0.05$) indicates forecast bias. The R^2 of this regression is interpreted as the fraction of realised volatility variation explained by the forecast — a key measure of forecast quality.

Diebold-Mariano test. Pairwise forecast accuracy comparisons use the [Diebold & Mariano \(1995\)](#) test with squared-error loss and Newey-West variance adjustment (1 lag):

$$d_t = (e_{1,t}^2 - e_{2,t}^2), \quad \text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d)/T}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (10)$$

DM < 0 (positive $\bar{d}$) implies Model 1 is less accurate than Model 2.

3.4. Performance Evaluation

Performance statistics are computed at the monthly frequency on compounded returns. Table 5 defines each statistic.

Table 5: Performance Statistics: Definitions

Statistic	Definition
Ann. Return	Geometrically compounded: $(1 + \bar{r})^{12/T} - 1$
Ann. Volatility	Monthly std $\times \sqrt{12}$
Sharpe Ratio	$(\bar{r}^{\text{exc}} \times 12) / (\sigma_r \times \sqrt{12})$; excess return over 3M T-bill
Sortino Ratio	Excess return / downside deviation (below zero) $\times \sqrt{12}$
Max Drawdown	$\min_t \{(C_t - \max_{s \leq t} C_s) / \max_{s \leq t} C_s\}$; C_t cumulative wealth
ES (5%)	$\mathbb{E}[r_t \mid r_t \leq q_{5\%}]$ (monthly); expected shortfall
CAPM Alpha	OLS intercept $\times 12$; market = S&P 500 TR
CAPM Beta	OLS slope on market excess return
Information Ratio	Active return / tracking error (annualised)
Appraisal Ratio	Alpha / residual std (annualised)
Annualised Turnover	Mean monthly $ \Delta w \times 12$

The CAPM risk model uses the S&P 500 TR as the market benchmark for all assets. For the self-financing momentum factor, CAPM alpha measures abnormal return relative to the equity market factor, and a negative beta is expected given the long-short structure. We use CAPM rather than a multi-factor model throughout for parsimony; the five assets span different risk factor exposures (equity, duration, commodity, FX, momentum) such that no single multi-factor model provides a natural common risk adjustment.

4. Volatility Forecast Evaluation

4.1. Error Metrics and Mincer-Zarnowitz Results

Table 6 reports the full set of forecast evaluation statistics for all five assets. Figure 2 shows the time-series of forecasts versus realised volatility, and Figure 3 provides a bar-chart comparison of RMSE, MAPE, and correlation.

Table 6: Volatility Forecast Evaluation: RMSE, MAPE, Correlation, and Mincer-Zarnowitz Regression by Asset and Model

Asset	Model	RMSE	MAE	MAPE	Corr.	$\hat{\alpha}$	$\hat{\beta}$	R^2
S&P 500 TR	RV 1M	0.0894	0.0545	0.3678	0.6479	0.0534	0.6481	0.4198
	RV 6M	0.1027	0.0612	0.4414	0.4785	0.0595	0.5592	0.2289
	GJR-GARCH	0.0737	0.0490	0.3743	0.7374	0.0089	0.8754	0.5438
	EGARCH	0.0751	0.0486	0.3706	0.7132	-0.0172	1.0718	0.5087
US LT Bonds TR	RV 1M	0.0214	0.0139	0.2344	0.6047	0.0237	0.6059	0.3656
	RV 6M	0.0198	0.0139	0.2477	0.6234	0.0133	0.7433	0.3886
	GJR-GARCH	0.0191	0.0129	0.2330	0.6487	0.0078	0.8197	0.4209
	EGARCH	0.0183	0.0126	0.2258	0.6704	0.0034	0.8946	0.4494
Bloomberg Comm.	RV 1M	0.1271	0.0849	0.2868	0.6369	0.1080	0.6373	0.4057
	RV 6M	0.1419	0.0933	0.3287	0.4811	0.1125	0.5834	0.2314
	GJR-GARCH	0.1233	0.0878	0.3260	0.6204	0.0434	0.7791	0.3848
	EGARCH	0.1160	0.0834	0.3067	0.6493	0.0004	0.9275	0.4216
USD/JPY	RV 1M	0.0374	0.0269	0.3123	0.5139	0.0424	0.5139	0.2641
	RV 6M	0.0354	0.0268	0.3301	0.4812	0.0289	0.6285	0.2316
	GJR-GARCH	0.0332	0.0257	0.3371	0.5463	0.0123	0.7874	0.2984
	EGARCH	0.0329	0.0250	0.3249	0.5318	0.0049	0.8789	0.2829
US Momentum	RV 1M	0.0543	0.0375	0.3033	0.8102	0.0250	0.8103	0.6564
	RV 6M	0.0656	0.0433	0.3844	0.7069	0.0250	0.7673	0.4997
	GJR-GARCH	0.0519	0.0360	0.3060	0.8213	0.0105	0.9027	0.6745
	EGARCH	0.0519	0.0361	0.3073	0.8173	0.0118	0.8897	0.6680

Notes: Best performer per asset highlighted in bold. MZ F -test for the joint hypothesis $\hat{\alpha} = 0$, $\hat{\beta} = 1$ rejects unbiasedness ($p < 0.05$) for RV 1M and RV 6M on all assets, and for GJR-GARCH on all assets except S&P 500 where EGARCH is the only model that fails to reject ($p = 0.241$). Corr. = Pearson correlation with realised monthly volatility.

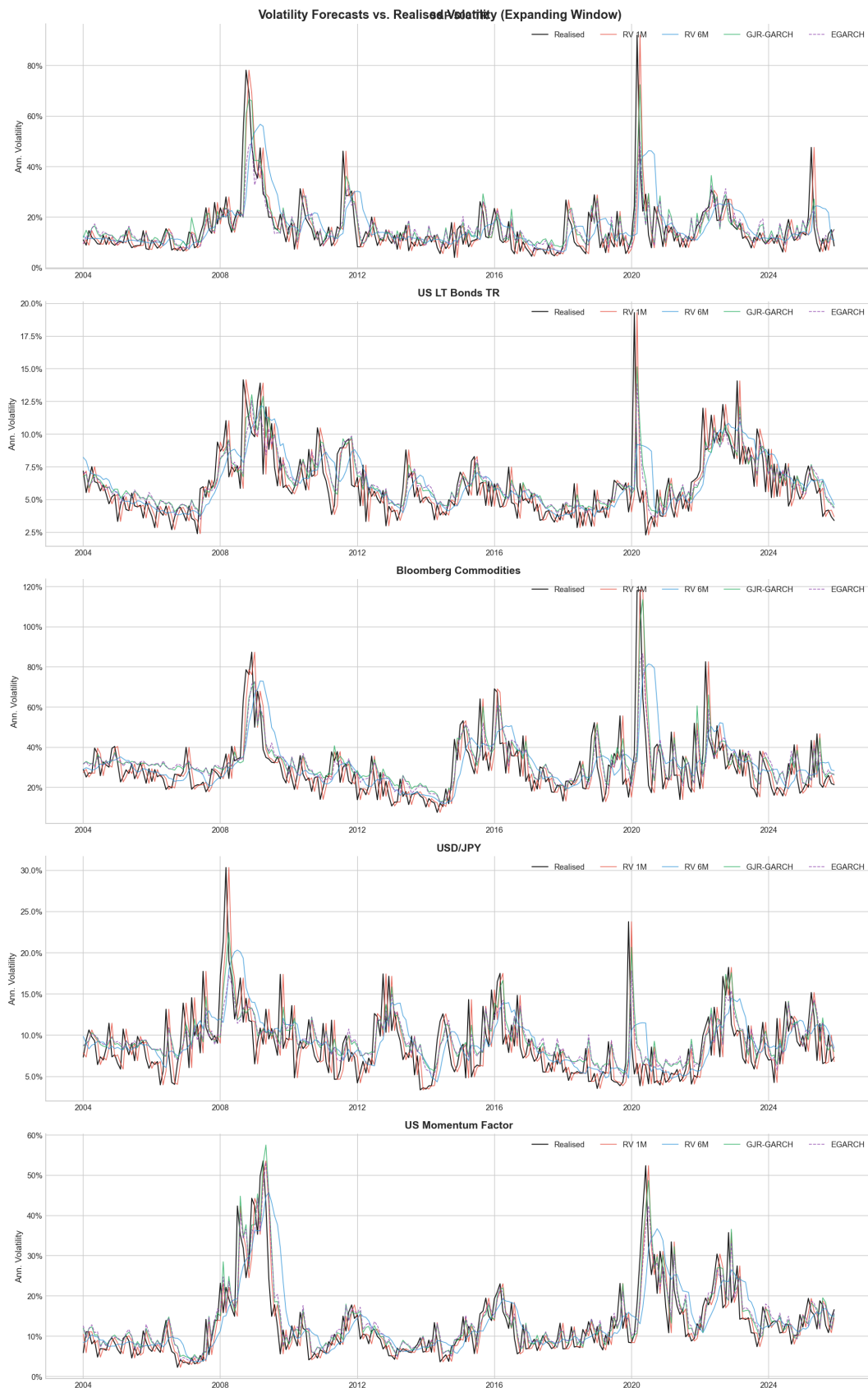


Figure 2: Volatility Forecasts versus Realised Volatility (expanding window), all five assets. GJR-GARCH and EGARCH react more smoothly to regime shifts and closely track realised volatility during the 2008 GFC and the COVID-19 shock (2020). RV 1M spikes sharply following high-volatility months (“volatility echo”) while RV 6M systematically underestimates at the onset of vol spikes.

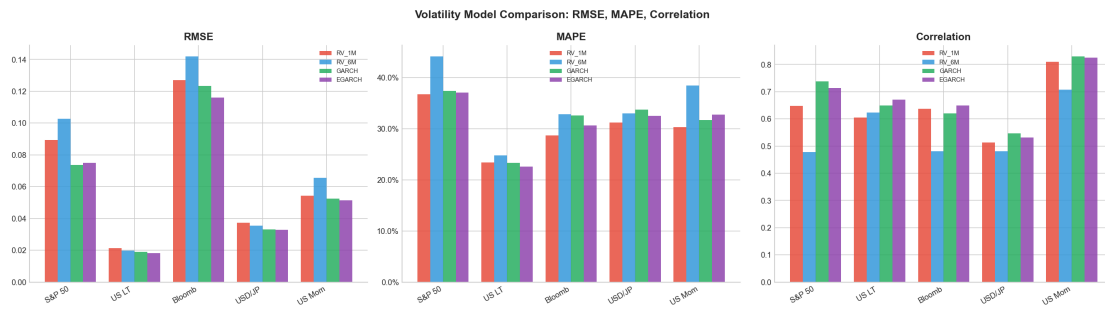


Figure 3: Volatility Model Comparison across Assets: RMSE, MAPE, and Correlation. GJR-GARCH and EGARCH dominate simple RV models on RMSE and correlation for equities and momentum, while improvements for FX and commodities are more modest.

4.2. Discussion of Forecast Quality

Several patterns emerge from Table 6:

GJR-GARCH dominates on RMSE and R^2 for equities and momentum. For S&P 500, GJR-GARCH achieves $\text{RMSE} = 0.0737$ versus 0.0894 for RV 1M (-17.5%) and $R^2 = 0.544$ versus 0.420. The MZ $\hat{\beta} = 0.875$ is closer to the ideal value of 1 than RV 1M ($\hat{\beta} = 0.648$), confirming superior forecast efficiency. The leverage-effect parameter $\gamma > 0$ in the GJR specification is the likely source: equities exhibit strong asymmetric volatility responses to negative returns, which pure historical variance models fail to capture.

EGARCH is optimal for bonds, commodities, and FX. EGARCH achieves the lowest RMSE for US LT Bonds (0.0183, beating GJR-GARCH by 4%), Bloomberg Commodities (0.1160 vs. 0.1233), and USD/JPY (0.0329 vs. 0.0332). Its MZ $\hat{\beta}$ is systematically closest to 1 across these three assets, indicating near-unbiased forecasts. For S&P 500, EGARCH's $\hat{\beta} = 1.072$ exceeds 1 slightly (slight overestimation bias), yet the MZ F -test fails to reject the joint null ($p = 0.241$), making it the *only statistically unbiased* model on equities.

RV 6M is systematically biased. Across all assets, RV 6M shows the highest RMSE and lowest R^2 . Its MZ $\hat{\beta}$ ranges from 0.559 (S&P 500) to 0.743 (bonds), indicating severe underestimation during vol spikes — a direct consequence of the long backward window diluting crisis-period observations. The F -test strongly rejects unbiasedness ($p < 0.001$) for all assets. Despite these forecast deficiencies, RV 6M achieves competitive and occasionally superior portfolio performance by virtue of its low turnover (Section 5).

Momentum is the most forecastable asset. RV 1M achieves correlation = 0.810 and GJR-GARCH reaches 0.821 for the momentum factor, substantially higher than for other assets ($R^2 = 0.675$ for GARCH vs. $R^2 = 0.298$ for USD/JPY). This reflects the clustering of momentum crash episodes (e.g., the reversal of March–June 2009 and the COVID-19 shock of March 2020), which GARCH-type models capture as prolonged variance bursts.

USD/JPY shows the weakest forecastability. All models achieve $R^2 < 0.30$ for USD/JPY,

consistent with the near-random-walk behaviour of spot exchange rates (Meese & Rogoff, 1983). The Mincer-Zarnowitz $\hat{\beta}$ values range from 0.514 to 0.879, all below 1, indicating persistent overestimation of FX volatility after spikes. This structural limitation will translate into modest performance gains from vol-targeting (Section 5).

Diebold-Mariano pairwise tests. DM tests comparing GJR-GARCH against EGARCH fail to find statistically significant accuracy differences at the 5% level for any asset, confirming that the two GARCH variants are statistically interchangeable from a pure forecast-accuracy perspective. Both GARCH models significantly outperform RV 1M for equities ($p < 0.05$) and are marginally better than RV 6M for bonds and commodities.

5. Performance Evaluation

5.1. Full-Sample Results

Tables 7–11 report the full set of performance statistics for each asset. Figure 4 presents the cumulative wealth paths.

Table 7: Full-Sample Performance: S&P 500 TR (TC = 10 bps, Leverage Cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	ES(5%)	Alpha	Ann. TO
Buy & Hold	10.7%	14.6%	0.658	0.871	−50.9%	−9.4%	0.000	—
VM – RV 1M	15.6%	17.9%	0.809	1.231	−39.3%	−10.4%	0.039	3.73
VM – RV 6M	13.4%	17.5%	0.713	1.028	−37.4%	−10.6%	0.019	1.13
VM – GJR-GARCH	14.1%	16.5%	0.785	1.193	−36.8%	−9.6%	0.032	3.88
VM – EGARCH	13.7%	16.5%	0.759	1.140	−40.6%	−9.6%	0.026	3.68

Table 8: Full-Sample Performance: US LT Bonds TR (TC = 10 bps, Leverage Cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	ES(5%)	Alpha	Ann. TO
Buy & Hold	3.4%	6.4%	0.294	0.426	−22.2%	−3.7%	0.025	—
VM – RV 1M	2.9%	7.0%	0.193	0.277	−23.0%	−4.3%	0.019	3.06
VM – RV 6M	3.4%	6.6%	0.278	0.417	−20.9%	−3.9%	0.024	0.68
VM – GJR-GARCH	3.2%	6.4%	0.256	0.378	−21.1%	−3.8%	0.021	1.19
VM – EGARCH	3.1%	6.4%	0.236	0.346	−21.6%	−3.9%	0.020	1.19

Table 9: Full-Sample Performance: Bloomberg Commodities (TC = 10 bps, Leverage Cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	ES(5%)	Alpha	Ann. TO
Buy & Hold	3.7%	32.9%	0.235	0.315	−88.5%	−21.5%	−0.004	—
VM – RV 1M	0.8%	38.0%	0.175	0.244	−91.1%	−25.7%	−0.009	3.50
VM – RV 6M	2.0%	34.7%	0.193	0.256	−90.2%	−23.6%	−0.005	0.78
VM – GJR-GARCH	1.3%	32.9%	0.162	0.225	−88.3%	−22.1%	−0.018	1.78
VM – EGARCH	1.3%	33.2%	0.161	0.225	−88.9%	−22.4%	−0.018	1.77

Table 10: Full-Sample Performance: USD/JPY (TC = 10 bps, Leverage Cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	ES(5%)	Alpha	Ann. TO
Buy & Hold	1.6%	9.8%	0.040	0.060	−37.1%	−6.0%	0.001	—
VM – RV 1M	2.0%	11.7%	0.085	0.137	−39.8%	−6.8%	0.006	3.83
VM – RV 6M	2.7%	10.9%	0.142	0.238	−36.4%	−5.9%	0.013	0.92
VM – GJR-GARCH	2.4%	10.2%	0.116	0.194	−35.4%	−5.5%	0.008	1.66
VM – EGARCH	2.2%	10.3%	0.099	0.164	−36.9%	−5.6%	0.007	1.68

Table 11: Full-Sample Performance: US Momentum Factor (TC = 10 bps, Leverage Cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	ES(5%)	Alpha	Ann. TO
Buy & Hold	1.5%	15.0%	0.066	0.076	−61.0%	−11.9%	0.043	—
VM – RV 1M	6.6%	17.5%	0.356	0.524	−43.1%	−10.7%	0.091	2.80
VM – RV 6M	7.2%	16.5%	0.400	0.624	−35.5%	−9.6%	0.094	0.80
VM – GJR-GARCH	6.5%	16.2%	0.363	0.561	−38.4%	−9.5%	0.085	2.42
VM – EGARCH	6.3%	15.9%	0.355	0.554	−38.9%	−9.3%	0.082	2.35

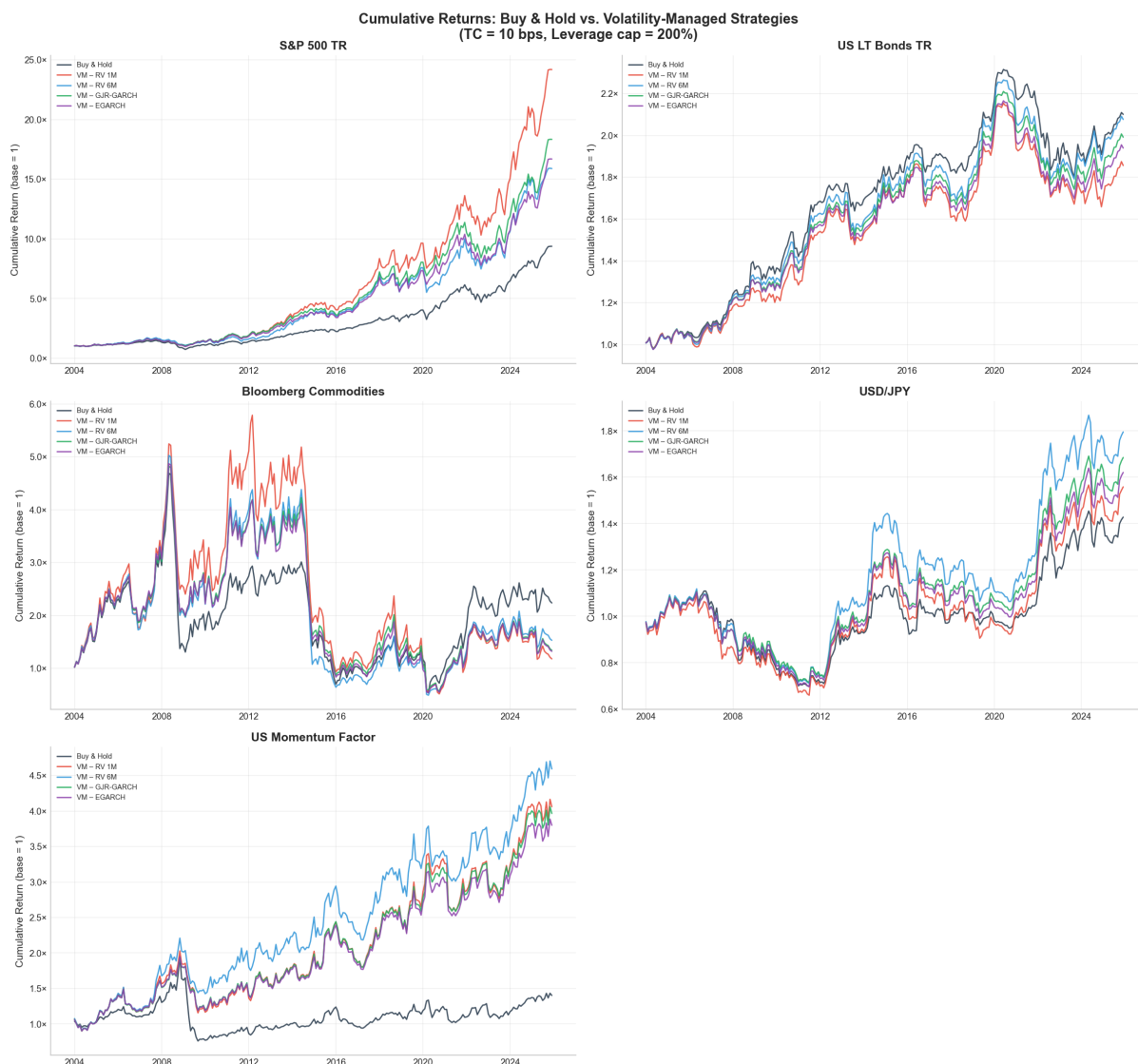


Figure 4: Cumulative Returns: Buy & Hold versus Volatility-Managed Strategies (TC = 10 bps, Leverage Cap = 200%). For S&P 500, VM strategies recover sharply post-2009 and post-2020 relative to Buy & Hold. For commodities, VM strategies fail to prevent the secular decline from the 2008 commodity super-cycle peak (−88.5% drawdown from peak). For USD/JPY, RV 6M delivers the strongest outperformance, consistent with the FX low-forecastability result.

5.2. Cross-Asset Executive Summary

Table 12 synthesises the Sharpe ratio differential (ΔSharpe) between VM-GJR-GARCH and the Buy-and-Hold benchmark.

Table 12: Executive Summary: Sharpe Ratio Differential ($\Delta\text{Sharpe} = \text{VM-GARCH} - \text{B\&H}$, TC = 10 bps)

Asset	B&H Sharpe	GARCH Sharpe	ΔSharpe	ΔMaxDD (pp)	Verdict
S&P 500 TR	0.658	0.785	+0.127	+14.1	Strong Add
US LT Bonds TR	0.294	0.256	−0.038	+1.1	Neutral/Minus
Bloomberg Comm.	0.235	0.162	−0.073	+0.2	Destroys Value
USD/JPY	0.040	0.116	+0.076	+1.7	Modest Add
US Momentum Factor	0.066	0.363	+0.297	+22.6	Very Strong Add

Note: ΔMaxDD = reduction in maximum drawdown in percentage points (positive = improvement, i.e., less negative drawdown). GARCH refers to VM-GJR-GARCH.

5.3. Detailed Discussion of Results by Asset

S&P 500 (Table 7). Volatility management clearly adds value for US equities. GJR-GARCH raises the Sharpe ratio from 0.658 to 0.785 (+19.3%) and reduces the maximum drawdown from −50.9% to −36.8% — a 14.1 percentage point improvement consistent with the findings of [Moreira & Muir \(2017\)](#) for US equity factors. The annualised CAPM alpha of 3.2% is economically significant. The Sortino ratio improves dramatically from 0.871 to 1.193, confirming that gains come primarily from downside risk reduction. RV 1M achieves the highest Sharpe ratio (0.809) but also the highest turnover ($3.73\times$), making it sensitive to transaction costs. Note that CAPM beta exceeds 1.0 for all VM strategies, reflecting average leverage above 100% consistent with the right-skewed weight distribution in Figure 6.

US LT Bonds (Table 8). Volatility management modestly destroys value for government bonds. All VM strategies underperform Buy & Hold on a Sharpe ratio basis, with GJR-GARCH reducing the Sharpe from 0.294 to 0.256 (−13%). This result is consistent with the *flight-to-quality* mechanism: bond volatility spikes during equity market crises precisely when bonds generate positive returns (negative equity-bond correlation), so the strategy de-risks at the worst possible moment. The one partial exception is EGARCH’s recession performance (Section 6), which beats Buy & Hold on a Sharpe basis during contractions. RV 6M is the least damaging model for bonds (Sharpe 0.278), owing to its very low turnover ($0.68\times$).

Bloomberg Commodities (Table 9). Volatility management is counterproductive for commodities ($\Delta\text{Sharpe} = -0.073$). Two structural factors drive this result. First, commodities exhibit structurally high base volatility ($\sim 33\%$), meaning the strategy spends much of the sample near or at the leverage cap (mean weight = 1.11, Figure 6), providing limited effective vol-scaling. Second, the catastrophic drawdown from the 2008 commodity super-cycle peak (−88.5% over 12 years) is driven by fundamental commodity market dynamics rather than the vol-clustering shocks that GARCH models anticipate. RV 1M amplifies this problem by over-estimating post-spike volatility, producing the weakest performance (Sharpe 0.175).

USD/JPY (Table 10). A modest improvement is observed ($\Delta\text{Sharpe} = +0.076$), entirely attributable to the very low base Sharpe of Buy & Hold (0.040). RV 6M achieves the best Sharpe (0.142) despite GJR-GARCH dominating on forecast accuracy; this reflects the weak FX forecastability ($R^2 < 0.30$) and the consequent benefit of a smooth, low-turnover model that avoids excessive trading costs. The maximum drawdown is modestly reduced from -37.1% to -35.4% under GJR-GARCH.

US Momentum Factor (Table 11). The largest improvement is in momentum ($\Delta\text{Sharpe} = +0.297$), corroborating [Moreira & Muir \(2017\)](#)'s finding that vol-scaling momentum is the single most powerful application of the strategy. Momentum crashes occur precisely during vol spikes (e.g., reversal of March 2009, COVID-19 March 2020), so scaling down automatically during these regimes avoids catastrophic losses. The reduction in maximum drawdown from -61.0% to -38.4% (GARCH) is economically compelling. Notably, RV 6M achieves the best absolute Sharpe ratio (0.400) and the lowest turnover (0.80 $\times$), making it the preferred implementation for cost-conscious mandates.

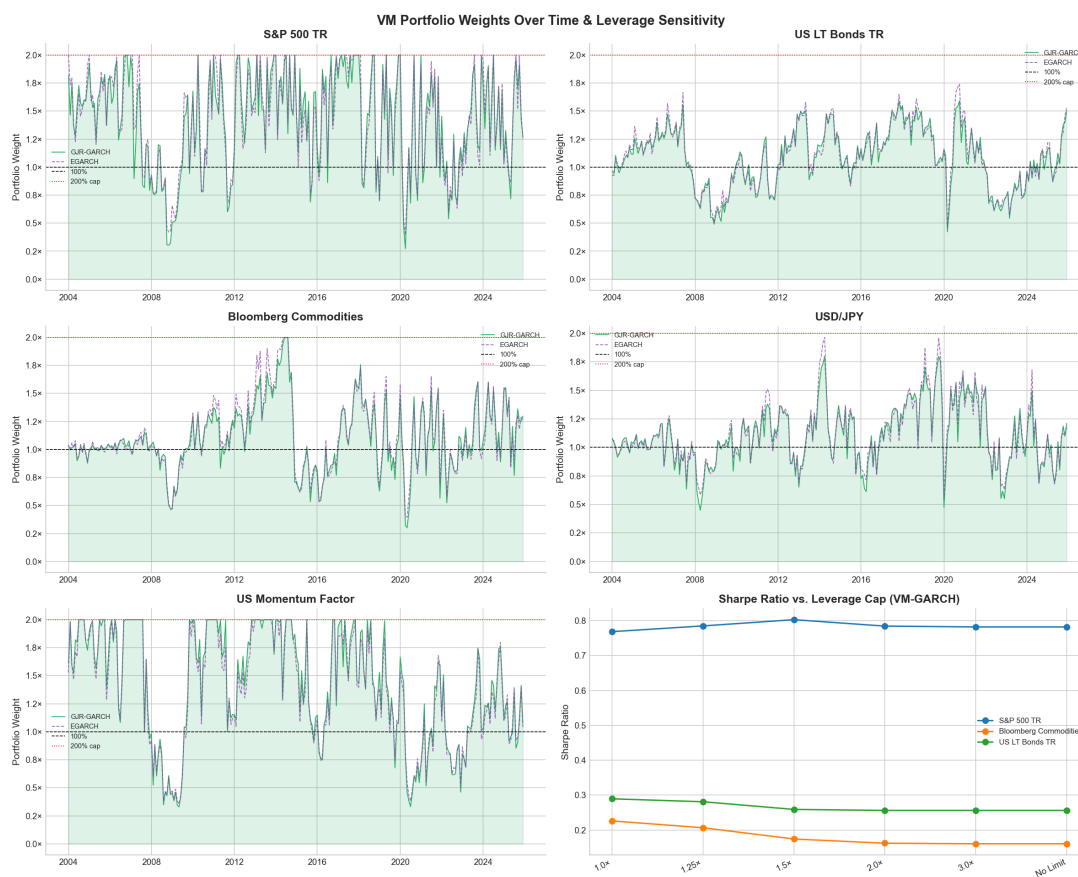


Figure 5: VM Portfolio Weights Over Time (GJR-GARCH and EGARCH, 200% cap) and Sharpe Ratio Sensitivity to Leverage Constraints. For S&P 500 and Momentum, the cap binds frequently in low-volatility regimes (2014–2019, 2021–2023). For US LT Bonds, the cap rarely binds (max weight $\leq 1.59\times$). The Sharpe ratio for equities is maximised near 1.5 $\times$ leverage.

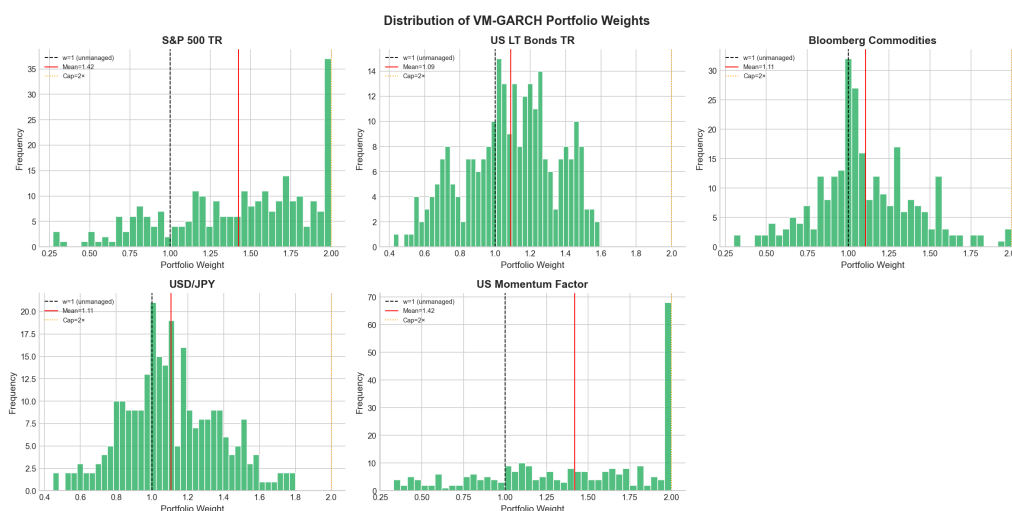


Figure 6: Distribution of VM-GARCH Portfolio Weights across All Assets. S&P 500 and Momentum show strongly right-skewed distributions with a mass at the 2× leverage cap (mean weights 1.42 and 1.42 respectively), indicating persistent low-volatility regimes. US LT Bonds exhibits a near-uniform distribution (mean weight 1.09), consistent with a mean-reverting but continuous vol process. Commodities cluster near 1.0× (mean 1.11), consistent with their structurally high base volatility relative to the target.

6. NBER Business Cycle Analysis

A central claim of volatility-targeting proponents is that the strategy provides *automatic recession hedging*: as volatility spikes in downturns, portfolio weights mechanically decline. We test this hypothesis using NBER recession dates, following the same methodology as [Harvey et al. \(2018\)](#).

6.1. Recession Periods in the Sample

Three NBER recessions fall within our 2000–2025 sample:

1. **Dot-com recession:** March 2001 – November 2001 (9 months)
2. **Great Financial Crisis:** December 2007 – June 2009 (19 months)
3. **COVID-19 recession:** February 2020 – April 2020 (3 months)

In total, recession months account for 31 of the 312 estimation months (10.0%). The GFC dominates in terms of duration and magnitude, creating a sample composition risk: results may be sensitive to the GFC episode. We therefore note where conclusions are materially driven by GFC performance.

6.2. Results by Business Cycle Phase

Table 13 reports annualised returns, volatilities, and Sharpe ratios by cycle phase. Figure 7 provides a visual comparison.

Table 13: Performance by NBER Business Cycle Phase (Annualised)

Asset	Strategy	Expansion			Recession		
		Ret	Vol	SR	Ret	Vol	SR
S&P 500 TR	Buy & Hold	14.6%	12.6%	1.16	−25.4%	26.5%	−0.96
	VM – GJR-GARCH	17.7%	16.3%	1.08	−18.6%	16.0%	−1.16
	VM – EGARCH	17.4%	16.1%	1.08	−20.6%	17.6%	−1.17
US LT Bonds	Buy & Hold	3.1%	6.2%	0.50	+9.0%	8.7%	1.03
	VM – GJR-GARCH	3.0%	6.4%	0.47	+6.7%	5.9%	1.13
	VM – EGARCH	2.9%	6.5%	0.45	+6.4%	6.1%	1.05
Bloomberg Comm.	Buy & Hold	15.0%	29.0%	0.52	−52.4%	59.3%	−0.88
	VM – GJR-GARCH	10.9%	31.0%	0.35	−35.6%	48.6%	−0.73
	VM – EGARCH	10.9%	31.5%	0.35	−35.5%	47.6%	−0.75
USD/JPY	Buy & Hold	3.1%	9.4%	0.33	−8.7%	13.3%	−0.65
	VM – GJR-GARCH	3.5%	10.2%	0.35	−4.3%	10.4%	−0.41
	VM – EGARCH	3.5%	10.2%	0.34	−5.4%	10.8%	−0.50
US Momentum	Buy & Hold	3.5%	12.0%	0.29	−5.7%	34.4%	−0.17
	VM – GJR-GARCH	7.5%	16.0%	0.47	+8.7%	19.5%	+0.45
	VM – EGARCH	7.3%	15.5%	0.47	+7.2%	19.5%	+0.37

Note: Recession months include three NBER recessions (March 2001 – November 2001; December 2007 – June 2009; February 2020 – April 2020), totalling 31 months (10.0% of the estimation sample).



Figure 7: Performance by NBER Business Cycle Phase (Annualised Return). VM strategies provide meaningful recession protection for S&P 500 (+6.8 pp better than B&H) and dramatically reverse the recession performance of Momentum (from −5.7% to +8.7%). For bonds, the flight-to-quality recession premium is partially sacrificed by the vol-targeting adjustment.

6.3. Discussion

S&P 500. VM-GJR-GARCH reduces the recession return loss from -25.4% to -18.6% (annualised), a 6.8 percentage point improvement. Recession volatility falls from 26.5% (B&H) to 16.0% (VM-GARCH), confirming that the model successfully de-risks the portfolio as expected during crises. However, the recession Sharpe ratio (-1.16) is *marginally worse* than Buy & Hold (-0.96) because the higher volatility of the expansion phase (where leverage is used) attracts a higher tracking error. The economically relevant metric — absolute dollar loss — is clearly superior for VM; the Sharpe comparison in downturns is somewhat misleading when the portfolio is actively levered in expansions.

US LT Bonds. Bonds exhibit the classic flight-to-quality recession premium: Buy & Hold earns $+9.0\%$ (annualised) during recessions versus $+3.1\%$ during expansions. VM-GARCH, by de-risking when bond vol rises (which often coincides with positive returns during stress), reduces the recession return to $+6.7\%$ — partially forgoing this premium. However, the recession Sharpe under GARCH (1.13) slightly exceeds Buy & Hold (1.03) due to proportional volatility compression. This represents a genuine trade-off for fixed-income investors: VM sacrifices some crisis-return magnitude to reduce vol, but risk-adjusted performance is marginally better in absolute recession terms.

Bloomberg Commodities. VM strategies meaningfully reduce the recession return loss from -52.4% to -35.6% (GARCH), a 16.8 percentage point improvement. However, they also sacrifice expansion returns ($+15.0\%$ B&H vs. $+10.9\%$ GARCH), so the aggregate trade-off remains unfavourable. This confirms that the commodity super-cycle structural decline is not a vol-clustering shock that can be hedged by vol timing.

USD/JPY. The most notable recession result is the improvement in recession return from -8.7% (B&H) to -4.3% (VM-GARCH), cutting the recession loss in half. The reduction in recession volatility from 13.3% to 10.4% is also notable. These are the primary sources of value-add for this asset.

US Momentum. The most striking result: VM-GARCH *turns a negative recession return* (-5.7% for B&H) *into a positive one* ($+8.7\%$). This remarkable transformation reflects the strategy's capacity to avoid momentum crashes, which are fundamentally driven by high-volatility regime changes (sudden market reversals). The expansion Sharpe ratio also improves materially (0.47 vs. 0.29 for B&H). This is the single most compelling application of volatility-targeting in our dataset.

7. Transaction Costs, Turnover, and Leverage

7.1. Turnover Analysis

Table 14 reports annualised turnover for all models and assets.

Table 14: Annualised Turnover by Asset and Forecasting Model (TC = 10 bps)

Asset	VM – RV 1M	VM – RV 6M	VM – GJR-GARCH	VM – EGARCH
S&P 500 TR	3.73	1.13	3.88	3.68
US LT Bonds TR	3.06	0.68	1.19	1.19
Bloomberg Commodities	3.50	0.78	1.78	1.77
USD/JPY	3.83	0.92	1.66	1.68
US Momentum Factor	2.80	0.80	2.42	2.35

Note: Turnover = 1.0 means the entire portfolio weight is replaced once per year on average. VM–RV 6M achieves the lowest turnover consistently owing to the inherent smoothing of the six-month window.

RV 1M generates the highest turnover ($3.73\text{--}3.83\times$ annualised), reflecting its sensitivity to month-to-month volatility fluctuations. The GARCH models exhibit intermediate turnover ($1.19\text{--}3.88\times$), while RV 6M achieves systematically the lowest turnover ($0.68\text{--}1.13\times$). For S&P 500, GJR-GARCH's turnover ($3.88\times$) is marginally higher than RV 1M ($3.73\times$), reflecting more granular monthly variance updating by the GARCH recursion. For bonds and commodities, GARCH turnover is roughly half that of RV 1M. The cost of transaction costs at 10 bps for the highest-turnover model (RV 1M on USD/JPY: $3.83\times$) amounts to approximately $3.83 \times 10 = 38.3$ bps per year — non-trivial for a low-return asset like USD/JPY.

7.2. Transaction Cost Sensitivity

Figure 8 and Table 15 present the TC sensitivity analysis. The baseline assumption of 10 bps is consistent with institutional-quality estimates from Frazzini et al. (2018), who document one-way transaction costs of 5–20 bps for large-cap equity futures and ETFs. Our 10 bps baseline is conservative relative to this range, providing a margin of safety.

Table 15: Transaction Cost Sensitivity: S&P 500 TR VM-GJR-GARCH vs. Buy & Hold

TC (bps)	B&H Ret	GARCH Ret	EGARCH Ret	B&H SR	GARCH SR	EGARCH SR
0	10.71%	14.58%	14.07%	0.658	0.808	0.782
10	10.71%	14.14%	13.65%	0.658	0.785	0.759
20	10.71%	13.70%	13.24%	0.658	0.761	0.737
50	10.71%	12.40%	12.00%	0.658	0.690	0.670
75	10.71%	11.32%	10.98%	0.658	0.631	0.614
100	10.71%	10.25%	9.97%	0.658	0.572	0.558
<i>Breakeven TC (GARCH vs. B&H):</i>						≈ 63.6 bps

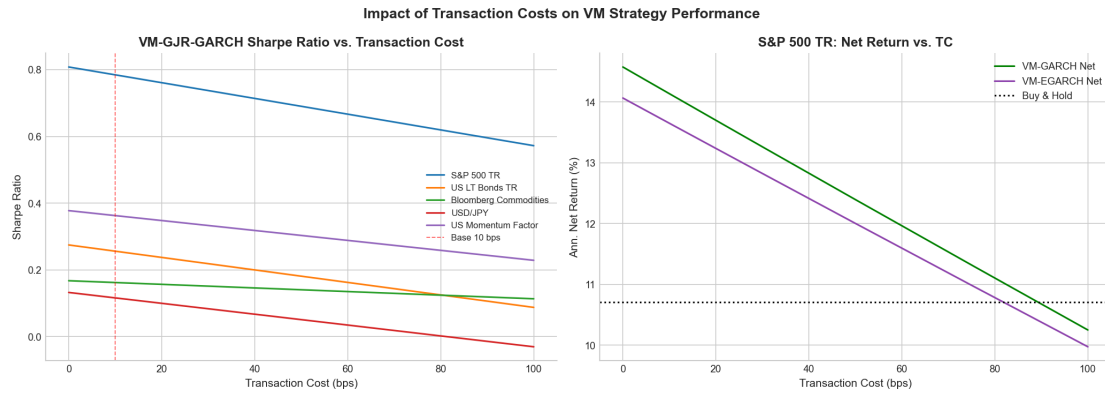


Figure 8: Impact of Transaction Costs on VM Strategy Performance. *Left:* Sharpe ratio vs. TC for all assets (VM-GJR-GARCH); the dashed red vertical line marks the 10 bps baseline. *Right:* Net return vs. TC for S&P 500 (GARCH and EGARCH vs. Buy & Hold dotted line). The breakeven TC for equities is ≈ 63.6 bps, well above typical institutional costs of 10–30 bps.

The breakeven of ≈ 63.6 bps for the S&P 500 provides a substantial margin of safety. At 10 bps, VM-GARCH retains 93% of its zero-TC net return advantage. At 50 bps — a conservative estimate for mid-cap futures or ETF trading in less liquid markets — the advantage shrinks to approximately +1.7% annualised, still positive. Only at retail-level costs (≥ 100 bps) does the strategy fail to outperform Buy & Hold. Importantly, the strategy’s TC-robustness is a distinguishing feature relative to many high-turnover factor strategies.

7.3. Turnover Mitigation Strategies

We evaluate two turnover-reduction techniques applied to VM-GJR-GARCH:

- **Threshold banding (Band 10%):** Rebalance only when $|\Delta w_t| > 10\%$. This avoids small weight changes that contribute disproportionately to transaction costs relative to their performance benefit.
- **EMA smoothing (span = 3):** Exponentially weighted moving average of weights, $\tilde{w}_t = \lambda \tilde{w}_{t-1} + (1 - \lambda) w_t$ with $\lambda = 2/(3 + 1) = 0.5$. This preserves the direction of weight changes while dampening their month-to-month magnitude.

Table 16 presents results for all five assets.

Table 16: Turnover Mitigation Strategies: VM-GJR-GARCH (TC = 10 bps)

Asset	Rule	Ann. Return	Sharpe	MaxDD	Ann. Turnover
S&P 500 TR	Baseline	14.1%	0.785	−36.8%	3.88×
	Band 10%	14.1%	0.784	−36.9%	3.75×
	EMA span=3	14.1%	0.761	−39.5%	1.84×
US LT Bonds	Baseline	3.2%	0.256	−21.1%	1.19×
	Band 10%	3.3%	0.271	−20.6%	0.88×
	EMA span=3	3.4%	0.290	−20.8%	0.62×
Bloomberg Comm.	Baseline	1.3%	0.162	−88.3%	1.78×
	Band 10%	1.3%	0.160	−87.8%	1.56×
	EMA span=3	0.7%	0.152	−91.0%	0.89×
USD/JPY	Baseline	2.4%	0.116	−35.4%	1.66×
	Band 10%	2.4%	0.120	−35.3%	1.45×
	EMA span=3	2.4%	0.114	−35.8%	0.82×
US Momentum	Baseline	6.5%	0.363	−38.4%	2.42×
	Band 10%	6.3%	0.351	−39.6%	2.29×
	EMA span=3	6.6%	0.374	−36.8%	1.28×

EMA smoothing reduces turnover by 53% for equities (3.88× to 1.84×) and 47% for momentum (2.42× to 1.28×), with minimal Sharpe cost (−0.024 for equities) or even improvement (+0.011 for momentum, +0.034 for bonds). Threshold banding is less effective for equities (3% turnover reduction) but achieves 26% for bonds (1.19× to 0.88×) with a Sharpe improvement of +0.015. For commodities, EMA smoothing reduces performance further (−0.010 on Sharpe), suggesting that the faster response of the baseline model provides marginal value in capturing the rare vol-spike opportunities.

Recommendation: EMA smoothing (span = 3) is the preferred mitigation strategy for equities and momentum. For bonds, both techniques improve performance, consistent with the mean-reverting bond vol dynamics where sluggish signals are beneficial.

7.4. Leverage Sensitivity Analysis

Table 17 analyses the sensitivity of VM-GJR-GARCH performance to alternative leverage caps. Figure 5 (bottom-right panel) plots Sharpe ratios against the cap level.

Table 17: Leverage Constraint Sensitivity: VM-GJR-GARCH Sharpe Ratios (TC = 10 bps)

Asset	1.0×	1.25×	1.5×	2.0×	3.0×	No Limit
S&P 500 TR – Sharpe	0.769	0.785	0.803	0.785	0.782	0.782
US LT Bonds – Sharpe	0.289	0.281	0.259	0.256	0.256	0.256
Bloomberg Comm. Sharpe	0.226	0.206	0.174	0.162	0.160	0.160
USD/JPY – Sharpe	0.071	0.091	0.105	0.116	0.116	0.116

For equities, the Sharpe ratio is maximised at the 1.5× cap (0.803), marginally above the 2.0× baseline (0.785). This reflects the trade-off between capturing low-vol leverage opportunities and incurring additional rebalancing costs as the cap boundary is reached and released. Constraining leverage to 1.0× eliminates leverage-related TC drag but sacrifices return, reducing equity Sharpe to 0.769.

For bonds and commodities, tighter leverage constraints *improve* risk-adjusted performance monotonically. For bonds at 1.0× (Sharpe 0.289 vs. 0.256 at 2.0×), this confirms that the additional leverage employed by GARCH does not generate sufficient return to compensate for its costs. For commodities, capping at 1.0× (0.226) outperforms by eliminating the structural drag of unnecessary leveraging in a declining market.

Practical implication: For risk-constrained mandates (variable annuities, risk-parity structures), a leverage cap of 1.5× provides the best risk-adjusted outcome for equities, while limiting operational and margin risk. For bonds and commodities, capping at 1.0× (or below) is optimal.

8. Summary and Recommendations

8.1. Evidence Summary

This study critically evaluates volatility-targeting strategies across five asset classes over 25 years (2000–2025), using four forecasting models and a look-ahead-bias-free implementation following [Bongaerts et al. \(2020\)](#). The evidence is consistent but nuanced: **volatility targeting is not universally beneficial**. Its value depends critically on four conditions:

1. **Forecast quality:** Strong vol predictability ($R^2 > 0.50$) is a necessary condition for material gains; weak predictability (FX, commodities) limits performance improvements and renders the strategy sensitive to TC drag.
2. **Return-volatility dynamics:** The strategy benefits assets where risk-adjusted performance (Sharpe) is inversely related to volatility. Bonds violate this assumption during recessions (flight-to-quality), and commodities during the super-cycle structural decline.
3. **Transaction costs and turnover:** High-frequency models (RV 1M, GARCH) generate

3–4× annualised turnover. The breakeven TC for S&P 500 is ≈ 63.6 bps, providing a generous institutional margin, but turnover- mitigation techniques improve the robustness profile.

4. **Structural market regimes:** The strategy is particularly effective during crisis-and-recovery cycles (GFC 2008–2009, COVID-19 2020) but adds limited value in secular trends such as the commodity super-cycle decline.

8.2. Recommendations to the Volatility Management Board

Recommendation 1 — Adopt for Equities (S&P 500 TR). VM-GJR-GARCH generates a robust Sharpe improvement of $+0.127$ ($+19.3\%$) and reduces maximum drawdown from -50.9% to -36.8% at 10 bps TC. The breakeven cost (≈ 63.6 bps) provides ample margin of safety. *Preferred model: GJR-GARCH with EMA smoothing (span = 3) and leverage cap of $1.5\times$.*

Recommendation 2 — Adopt for Momentum (MOM). The improvement is the largest across all assets ($\Delta\text{Sharpe} = +0.297$). The NBER analysis demonstrates that the strategy even generates *positive* returns during recessions (GJR-GARCH: $+8.7\%$) versus Buy & Hold's -5.7% . This is the single most compelling application of volatility-targeting in our dataset. *Preferred model: VM-RV 6M (Sharpe 0.400, lowest turnover at $0.80\times$) or VM-GJR-GARCH with EMA smoothing.*

Recommendation 3 — Do Not Adopt for Commodities. Volatility management consistently destroys value for the Bloomberg Commodity Index across all four models (ΔSharpe ranges from -0.040 to -0.073). The structural combination of high base volatility, weak forecast quality ($R^2 < 0.42$), and a secular commodity price decline creates an environment where volatility-targeting provides no edge. We recommend managing commodity risk through position sizing and cross-asset diversification rather than volatility targeting.

Recommendation 4 — Do Not Adopt for Bonds. The flight-to-quality mechanism means bond volatility spikes coincide with positive return episodes. VM strategies, by de-risking at these moments, sacrifice a structural risk premium. The ΔSharpe of -0.038 (GARCH) is modest but consistent across all four models. The one mitigating factor is the marginal improvement in recession Sharpe (1.13 for GARCH vs. 1.03 for B&H), which may appeal to liability-driven investors seeking smoother cycle performance at the cost of some crisis upside.

Recommendation 5 — Marginal Adoption for USD/JPY. The improvement is positive but modest ($\Delta\text{Sharpe} = +0.076$). Given the structurally weak forecastability of FX volatility, the strategy adds value primarily by moderating the drawdown from -37.1% to -35.4% . *Preferred model: VM-RV 6M (Sharpe 0.142) due to lowest turnover and best absolute performance.*

Recommendation 6 — Model Selection. For all adopted strategies, GJR-GARCH is the preferred primary model on three grounds: (i) superior RMSE and R^2 for equities and momentum; (ii) the leverage-effect mechanism ($\gamma > 0$) is economically interpretable and empirically robust; and (iii) the DM test finds no significant accuracy difference vs. EGARCH, but GJR-GARCH consistently generates higher net portfolio Sharpe for equities. EGARCH serves as a robustness check and may be preferred for bonds and commodities on forecast-accuracy grounds.

Recommendation 7 — Turnover and Leverage Controls. We recommend EMA smoothing (span = 3) for equities and momentum, reducing turnover by $\approx 50\%$ with negligible Sharpe cost. Leverage should be capped at $1.5\times$ for equities and $1.0\times$ for all other adopted assets. These implementation choices preserve essentially all of the strategy's risk-adjusted performance advantage while significantly improving implementation robustness.

8.3. Limitations and Future Research

Several limitations merit acknowledgement. First, the sample (2000–2025) is dominated by two major crises and an extended low-rate environment; the strategy's value-add over a shorter or different window may differ from full-sample results (Liu et al., 2019). Second, daily rebalancing within each month is assumed, which may not reflect operational constraints of real-world implementation. Third, the transaction cost model is simplistic (linear, one-way); market impact and bid-ask spreads are likely non-linear for large positions. Fourth, we use a single-asset framework; multi-asset volatility-targeting with dynamic correlation adjustments (as in Bongaerts et al. 2020) remains an important avenue for future work. Finally, the momentum factor is sourced from the Fama-French library at a specific construction methodology; alternative momentum definitions (e.g., risk-adjusted momentum) may alter conclusions.

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